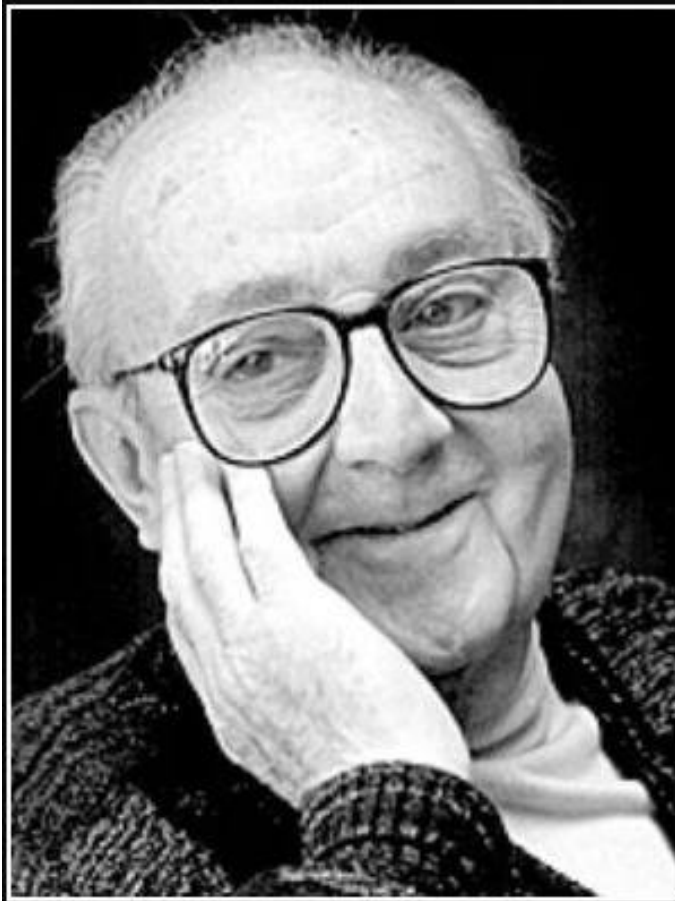


# Model Checking

# Announcements

- Chapter 6 and 7 today

# All models are wrong

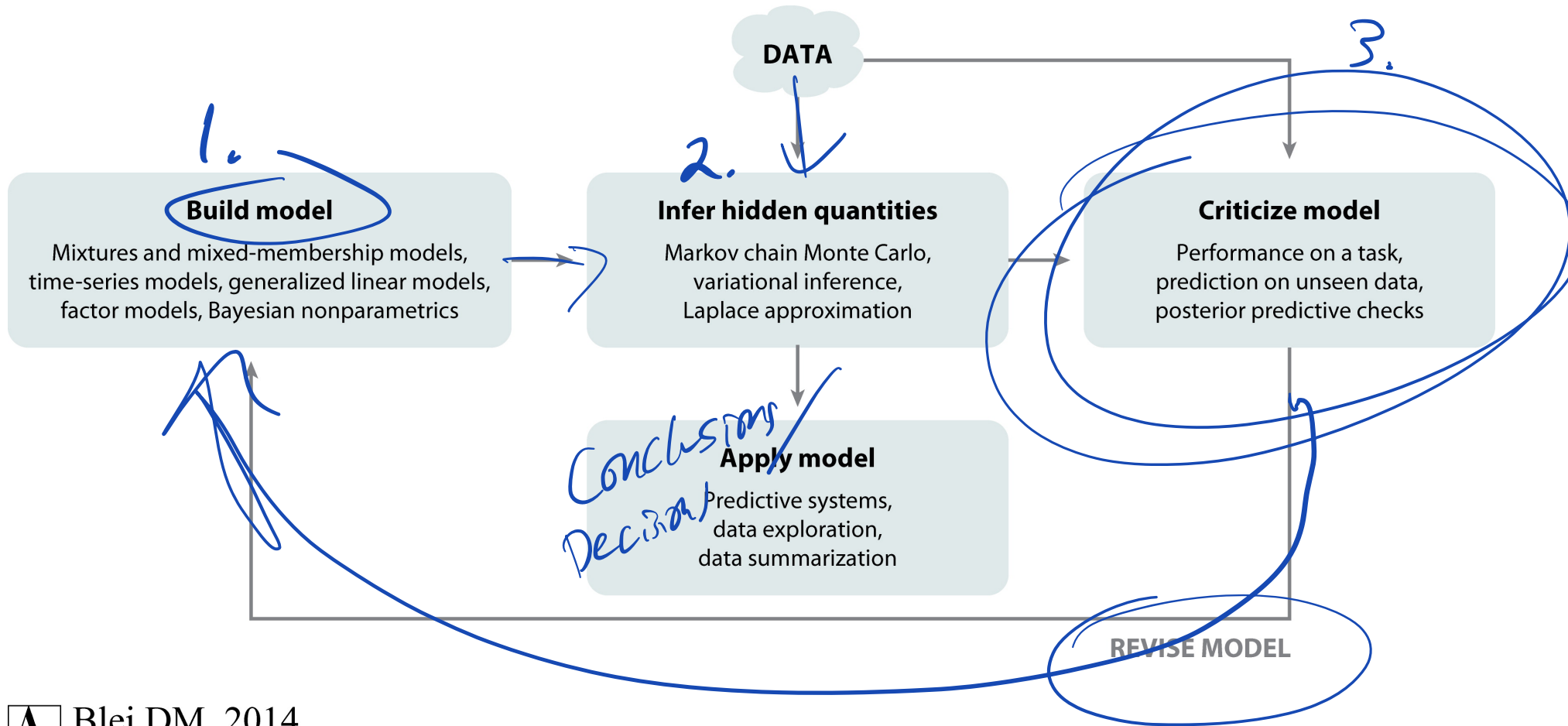


All models are wrong, but some are  
useful.

— *George E. P. Box* —

If the model is “wrong”, how can we improve it?

# Box's Loop



Blei DM. 2014.  
Annu. Rev. Stat. Appl. 1:203–32

# Posterior predictive model checking

- Let  $y_{\text{obs}}$  represent the observe data  $y_1, \dots, y_n$
- Let  $\tilde{y}$  represent  $n$  replicated (e.g fake) observations generated from the model
- $p(\tilde{y} \mid y_{\text{obs}}) = \int p(\tilde{y} \mid \theta)p(\theta \mid y_{\text{obs}})d\theta$
- Generate test quantity from  $t(\tilde{y})$  (summary statistic)
- Check if the simulated test quantities are similar to the observed test quantity,  $t(y_{\text{obs}})$

same says  
as actual  
data.

This can be done  
with Monte Carlo.

# Posterior predictive model checking

- If the model fits the data, then fake data generated under the model should look similar to the observed data
- Discrepancies can be due to model misfit or chance (or both!)
- Monte Carlo approach:
  1. sample  $\theta^{(s)} \sim p(\theta | Y = y_{\text{obs}})$
  2. sample  $\tilde{y}^{(s)} = (\tilde{y}_1^{(s)}, \dots, \tilde{y}_n^{(s)}) \sim \text{i.i.d. } p(y | \theta^{(s)})$ 
    - $\tilde{y}$  has same number of observations as  $y_{\text{obs}}$
  3. compute  $t^{(s)} = t(\tilde{y}^{(s)})$

$\theta \sim$

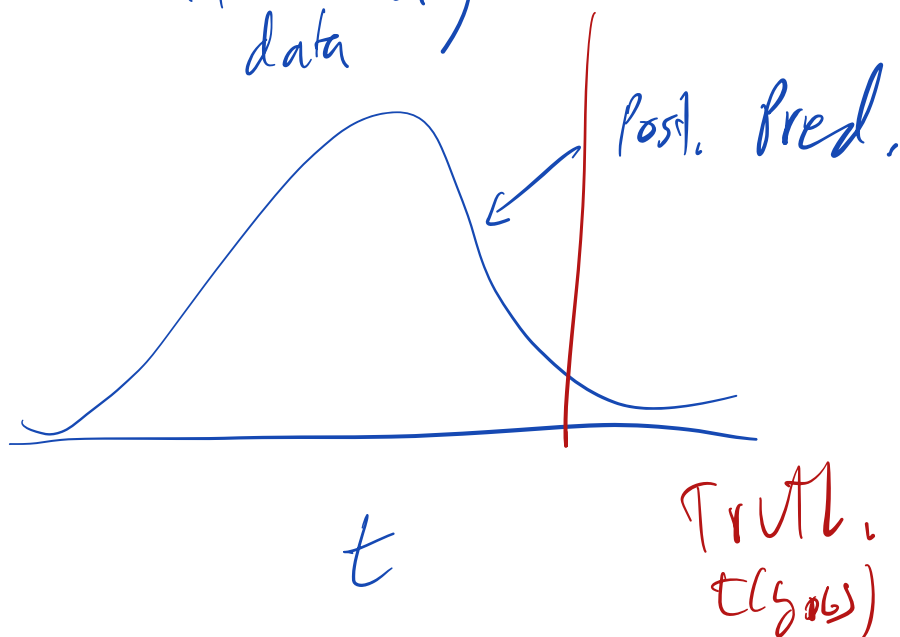
obs

"posterior  
predicted  
distribution  
of  $t$ "

Compare  $t_{obs}$  to  $\tilde{t}_{rep.}$

One stat  
(from real)  
data

Distribution



$(y) / y$

# A Bayesian Modeling Process (overview)

1. Propose a sampling model and prior distributions
2. Compute the posterior distribution, here
$$p(\theta \mid Y = y) \sim \text{Gamma}(a + y, \beta + \nu)$$
3. Simulate test statistics,  $T(\tilde{y})^{(s)}$  from the posterior predictive distribution
  - for  $s$  in  $1:\text{num\_fake\_data}$ 
    - Sample  $\theta^{(s)} \sim \text{Gamma}(a + y, b + \nu)$
    - Sample  $\tilde{y}^{(s)} \sim \text{i.i.d Pois}(\theta^{(s)})$  (same sample size as  $y_{obs}$ )
    - Compute  $T(\tilde{y}^{(s)})$



4. Compare the samples  $T(\tilde{y}^{(s)})$  to  $T(y_{\text{obs}})$
5. Identify any model misfit, go back to step 1 and repeat.

# Choosing Test Statistics

- Test quantity, or discrepancy measure,  $T(y, \theta)$  can also be a function of parameters
- A model can be inadequate for some purposes but adequate for others
- Avoid choosing sufficient statistics
- Statistics should be relevant to the scientific purpose
- Often measures a feature of the data not directly addressed by the probability model (ancillary statistics)
  - e.g. Ranks of the sample or correlation of residuals

# Speed of Light Example

- Newcomb measured the amount of time required for light to travel a distance of 7442 meters.
- 66 measurements were collected

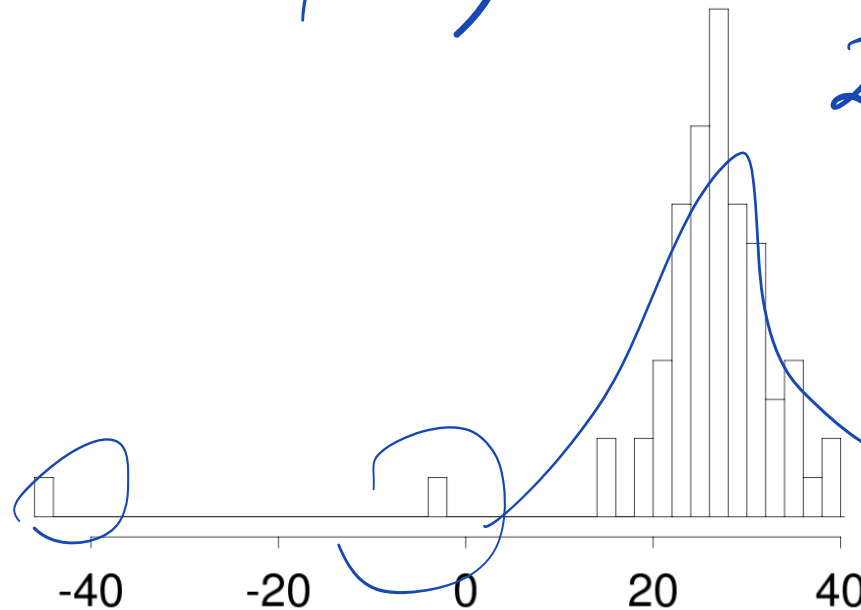
$$y_1, \dots, y_{66} \sim N(\mu, \sigma^2)$$

$$P(\mu, \sigma^2 | y_1, \dots, y_{66})$$

1.  $\mu^s, \sigma^s \sim \text{Posterior}$

2. Generate  
 $\tilde{y}_1, \dots, \tilde{y}_{66} \sim N(\mu^s, \sigma^s)$

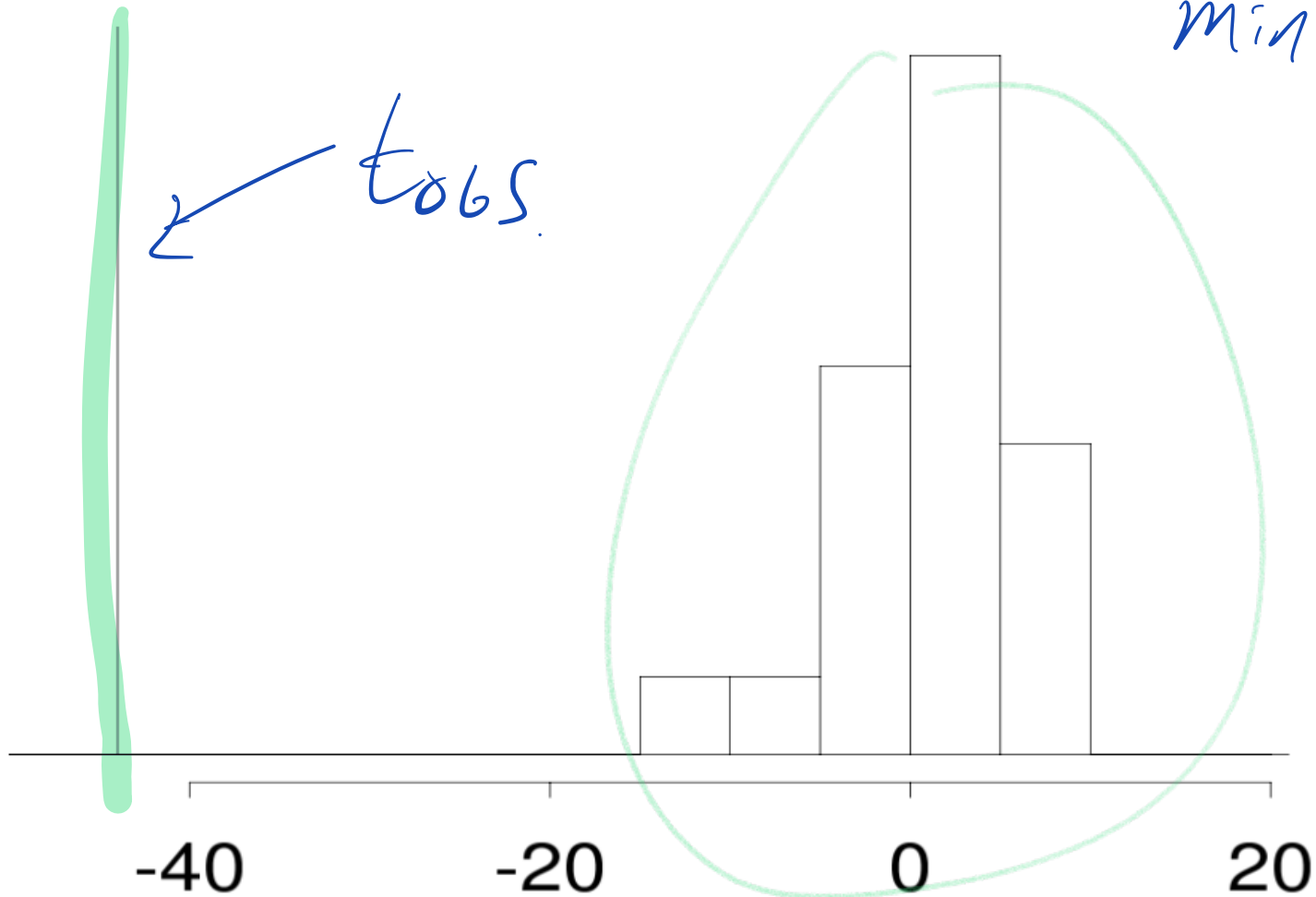
3. Compute  
 $t(\tilde{y}_1, \dots, \tilde{y}_{66})$



# Speed of Light ( $T = \text{min}$ )

$$t(\tilde{y}_1, \dots, \tilde{y}_{66}) =$$

$$\min(\tilde{y}_1^2, \dots, \tilde{y}_{66})$$

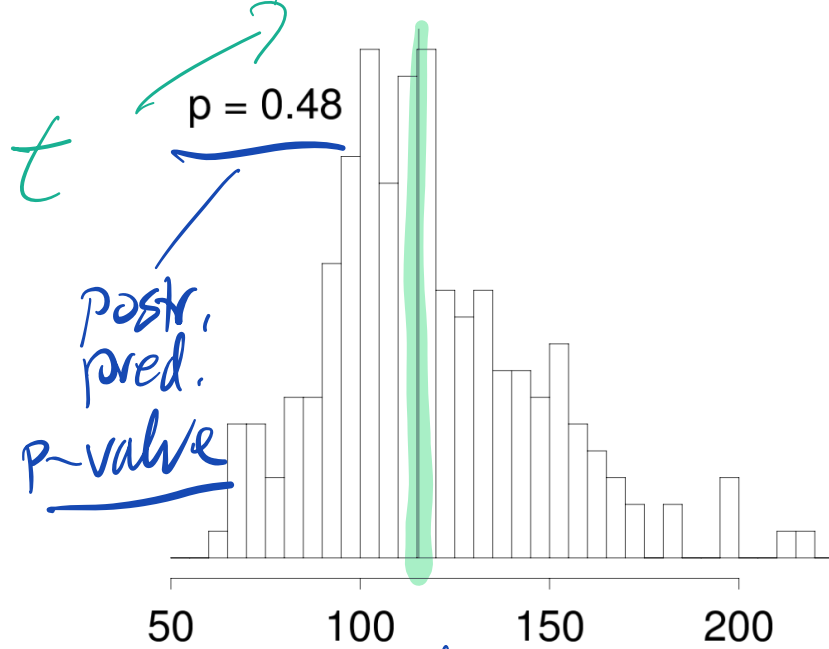


$$\hookrightarrow P(\min(\tilde{y}_1, \dots, \tilde{y}_{66}) | y_{obs})^{10}$$

# Speed of Light Example

$$T(y, \theta) =$$

$$P(\text{var}(\tilde{y}_1, \dots, \tilde{y}_{66}) | y_{\text{obs}})$$



post.  
pred.  
p-value

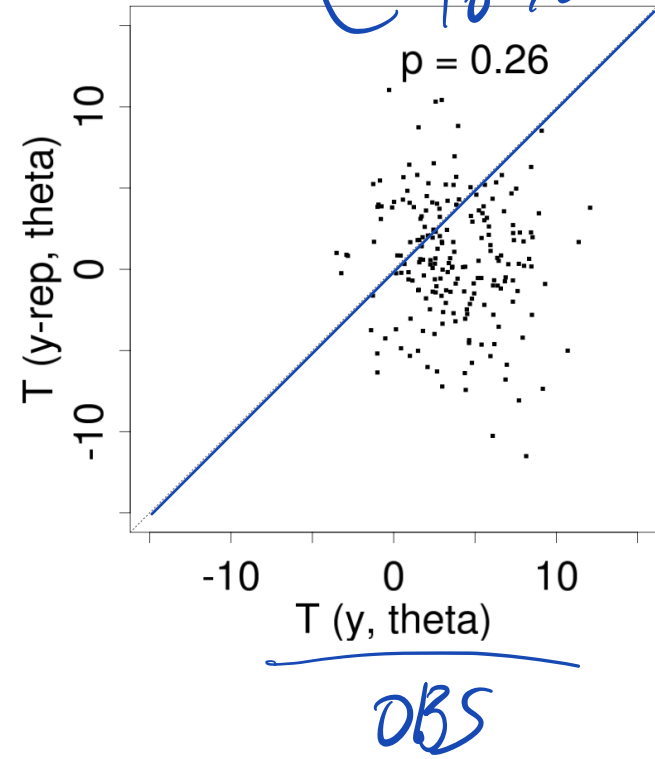
Good fit  
for variance.

$$\frac{|\tilde{y}_{(b)} - \mu|}{\sqrt{\text{var}(\tilde{y}_{(b)})}}$$

90% quantile

10%

Rep.



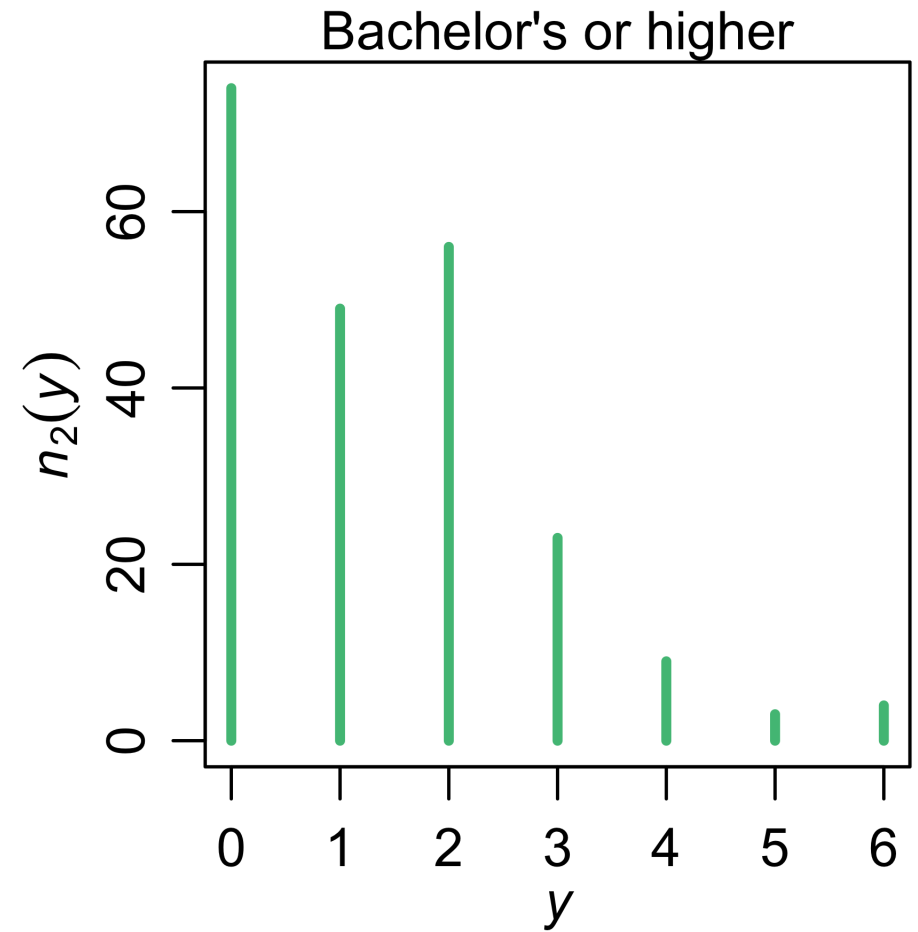
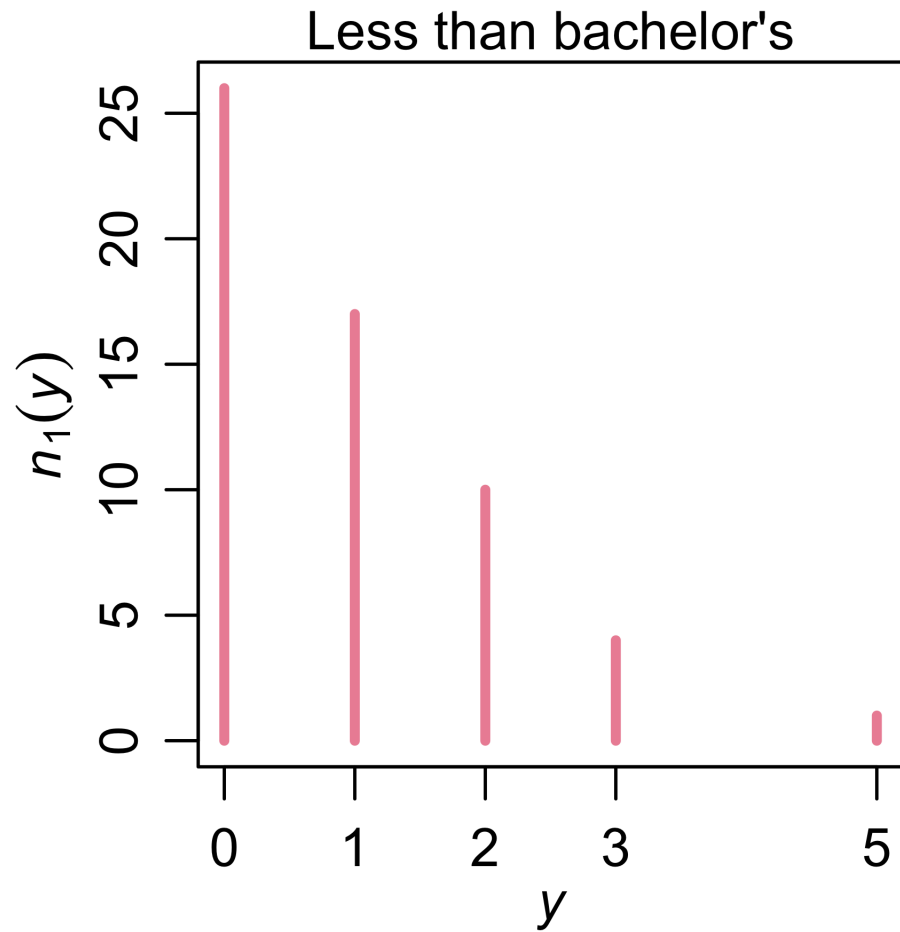
# Predictive Checks: an example

- In the 1990's there was a survey of 155 women, at least 40 years of age
- Recorded number of children and educational attainment
  - Bachelor's degree or higher ( $n_1 = 111$ )
  - Less than bachelor's degree ( $n_2 = 44$ )

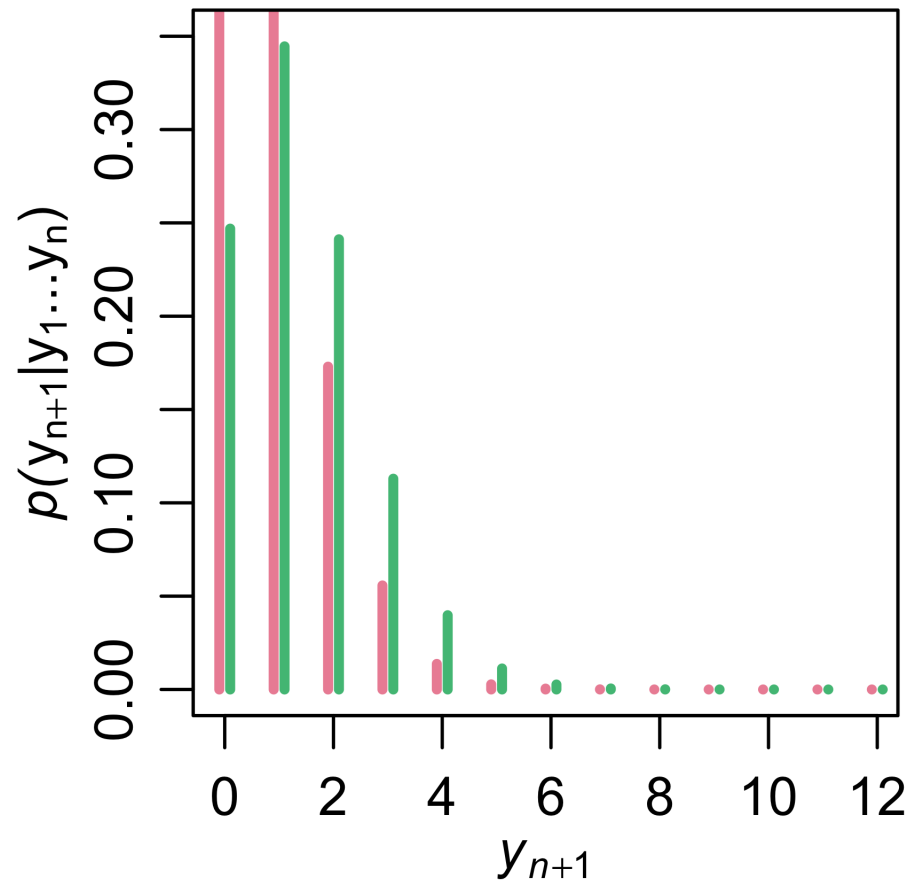
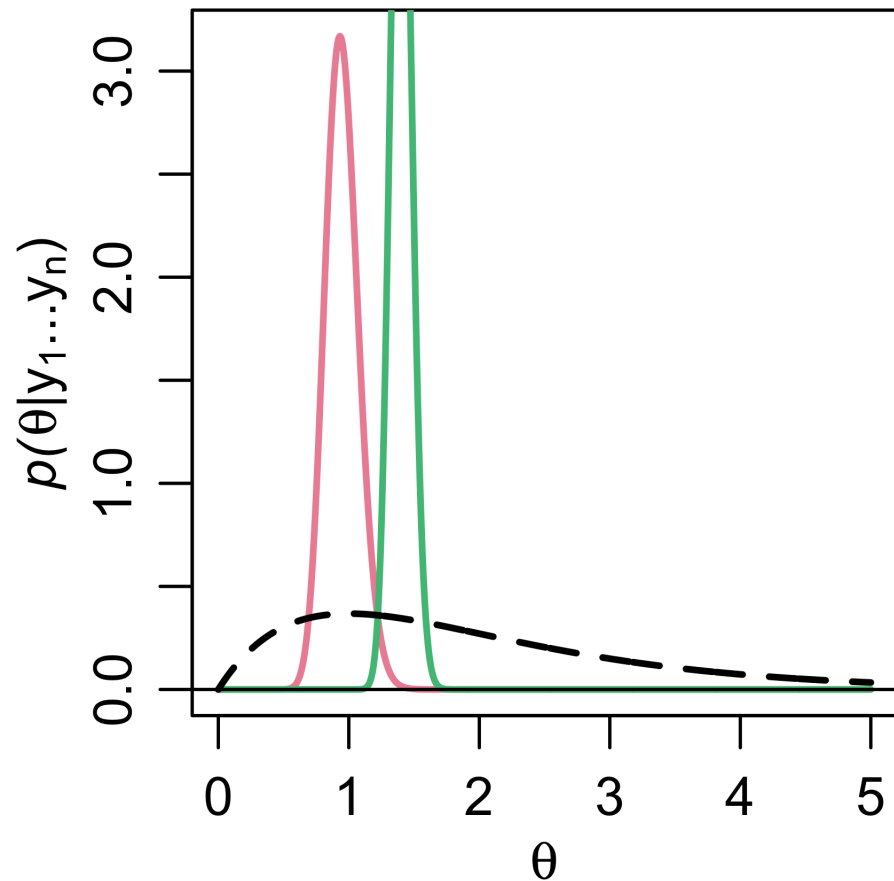
$$Y_{1,1} \dots, Y_{n_1,1} | \theta_1 \sim \text{i.i.d. Poisson}(\theta_1)$$

$$Y_{1,2} \dots, Y_{n_2,2} | \theta_2 \sim \text{i.i.d. Poisson}(\theta_2)$$

# PPCs example



# PPCs example





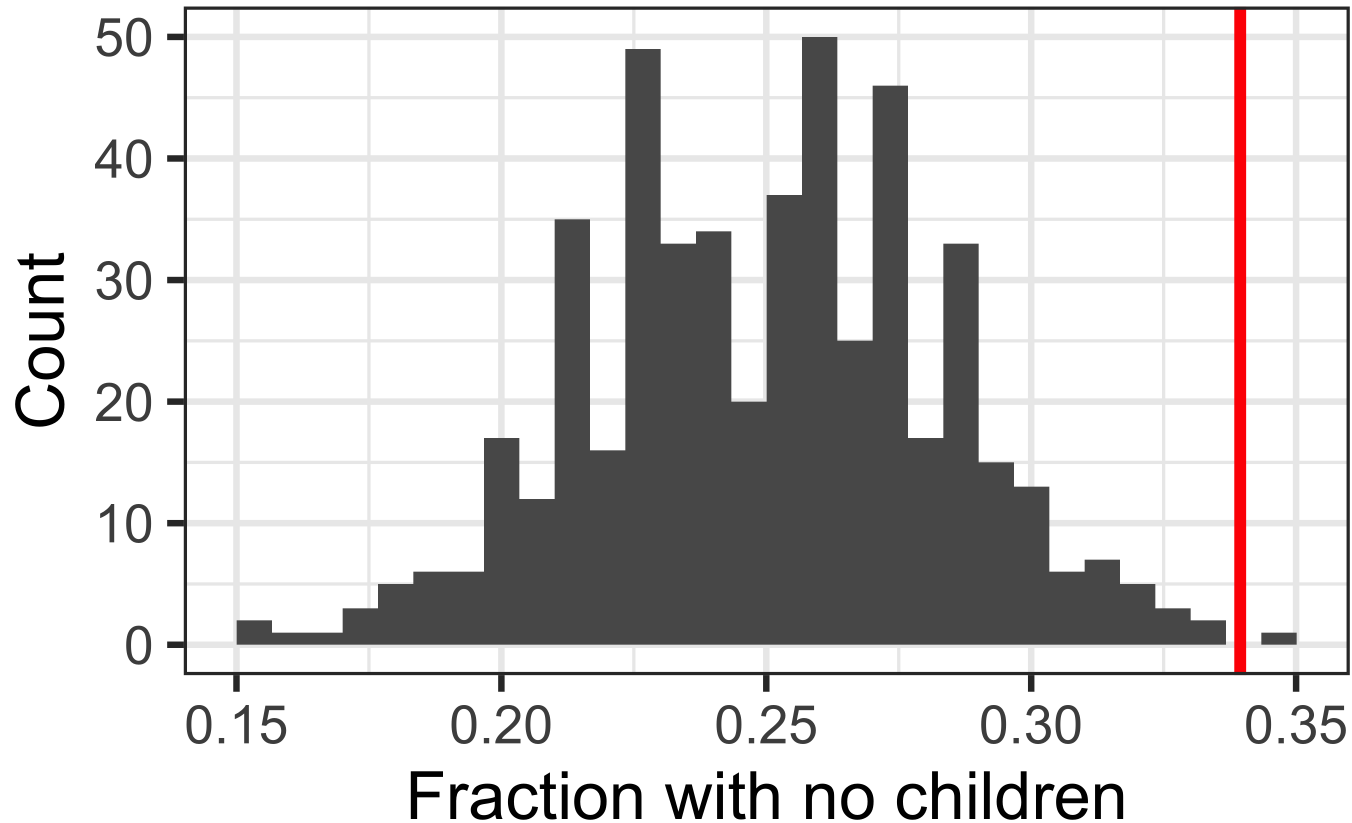
# PPCs example

- Let's check the model fit for the “without Bachelor's” group first
- Do  $S$  times:
  - sample  $n_2 = 44$  observations  $\tilde{y}$  from the posterior predictive distribution
- Let  $T(\tilde{y})$  be the fraction of women with no children

```
1  S <- 1000
2  t_s <- numeric(S)
3  for(s in 1:S){
4    theta_s <- rgamma(1, a, b) # whatever a and b are for my posterior
5    ytilde_s <- rpois(n=44, theta = theta_s)
6    t_s[s] <- mean(ytilde_s == 0) # compute test stat
7  }
8
9  ## then visualize histogram of t_s
```

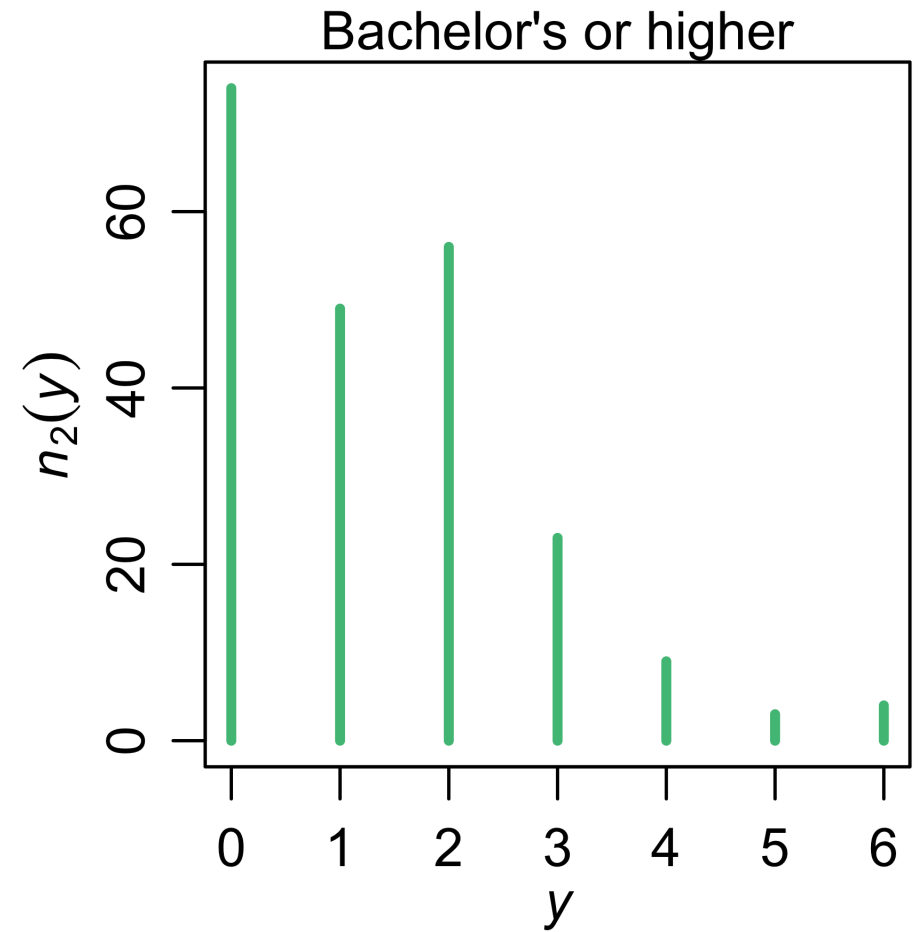
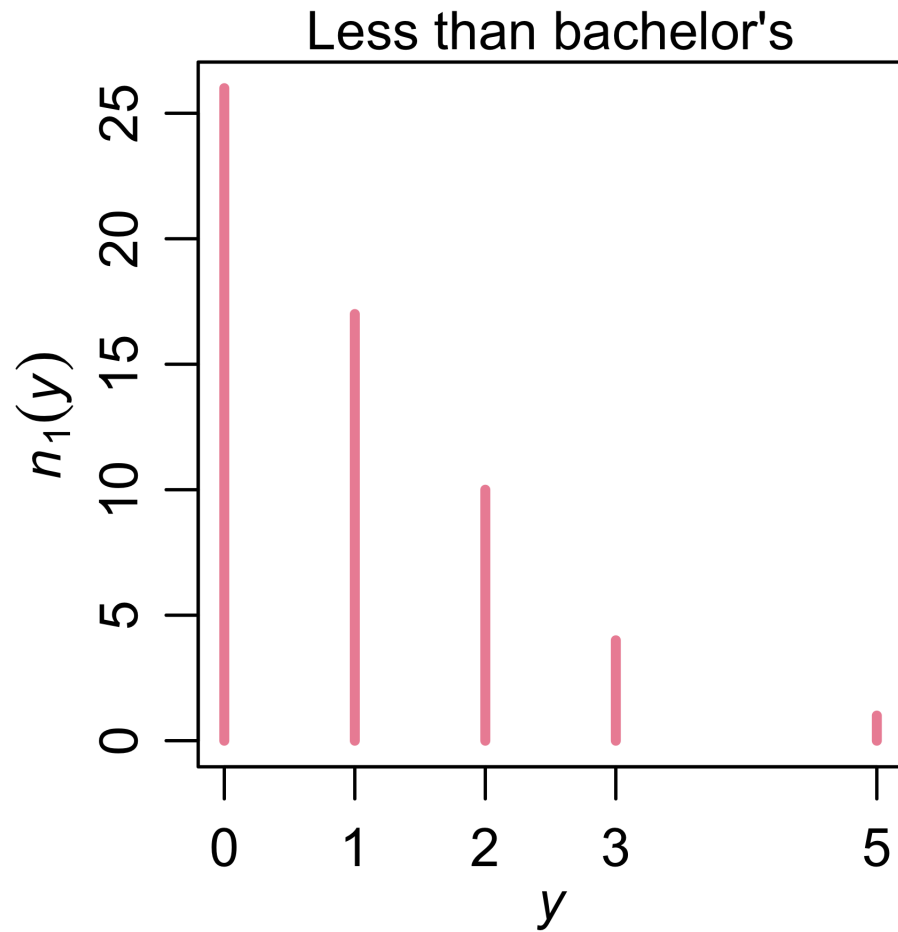
# PPCs example

Women without Bachelors degree



$$Pr((T^{rep} > T^{obs} \mid y_{obs}) = 0.002$$

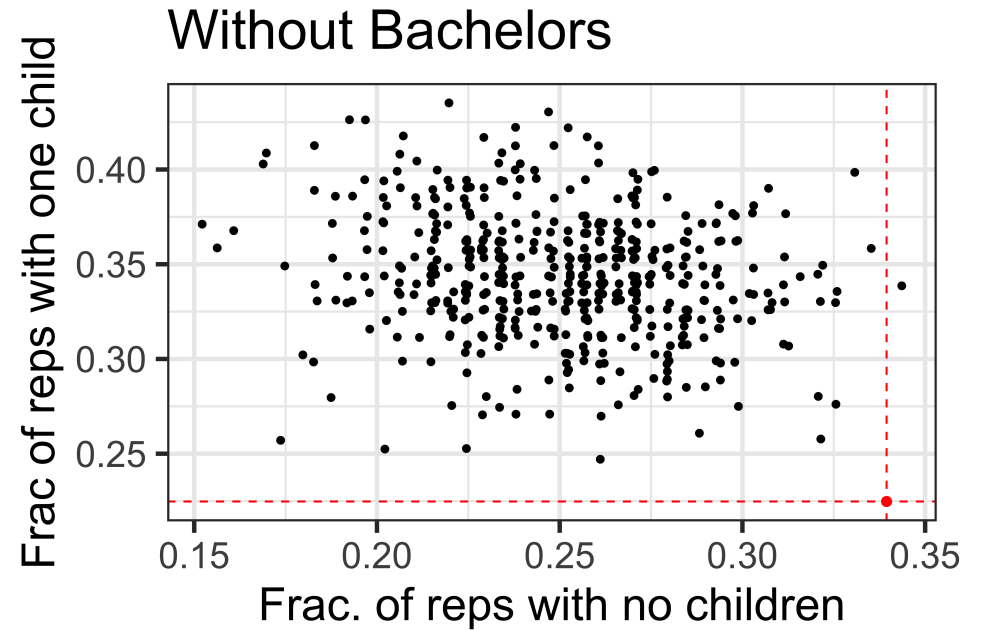
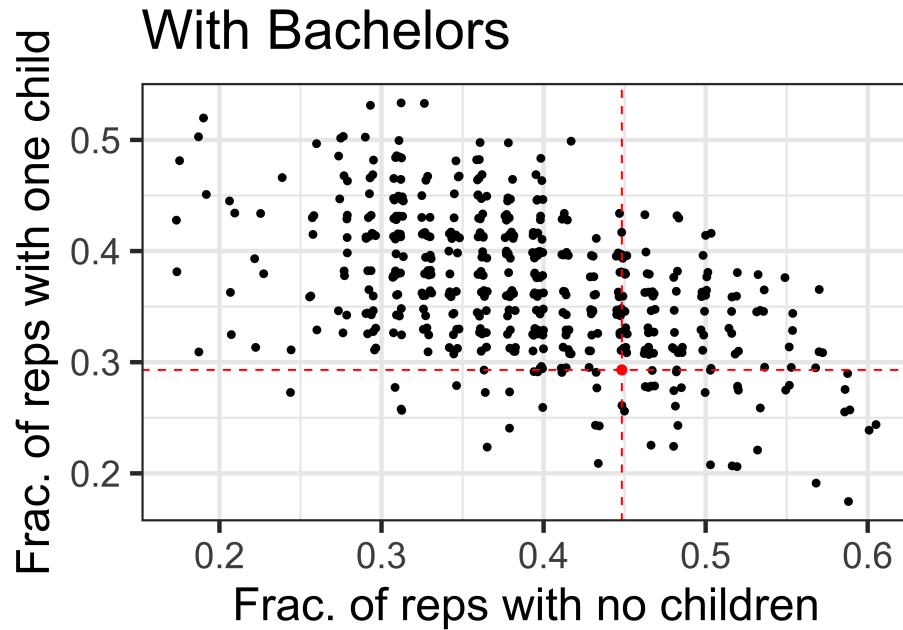
# PPCs example



# PPCs example

- Model checking both groups
- Look at fit for two different test statistics:
  - Fraction with no children
  - Fraction with one child

# Poisson example



# Predictive Distributions in Stan

```
1  ...
2  } model {
3    y1 ~ poisson(lambda1);
4    y2 ~ poisson(lambda2);
5  } generated quantities{
6
7    int y1rep[n1];
8    int y2rep[n2];
9    real log_lik[n1+n2];
10   for(i in 1:n1){
11     y1rep[i] = poisson_rng(lambda1);
12     log_lik[i] = poisson_lpmf(y1[i] | lambda1);
13   }
14   for(i in 1:n2){
15     y2rep[i] = poisson_rng(lambda2);
16     log_lik[n1+i] = poisson_lpmf(y2[i] | lambda2);
17   }
18 }
```

# Predictive Checks in Stan

Running MCMC with 4 sequential chains...

Chain 1 finished in 2.0 seconds.

Chain 2 finished in 2.1 seconds.

Chain 3 finished in 2.3 seconds.

Chain 4 finished in 2.0 seconds.

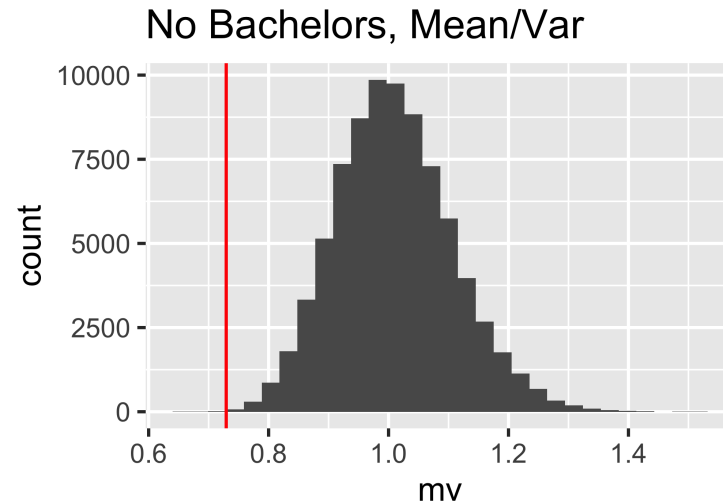
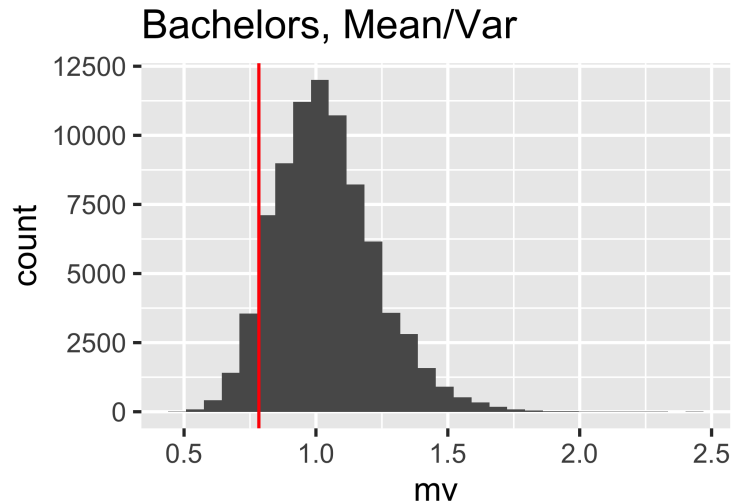
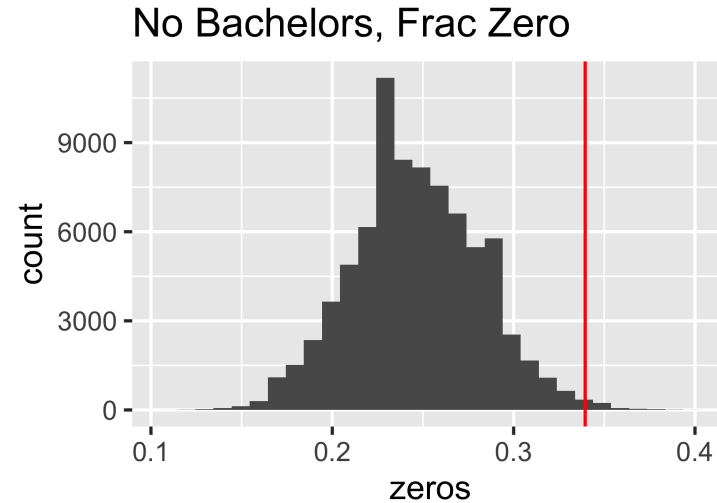
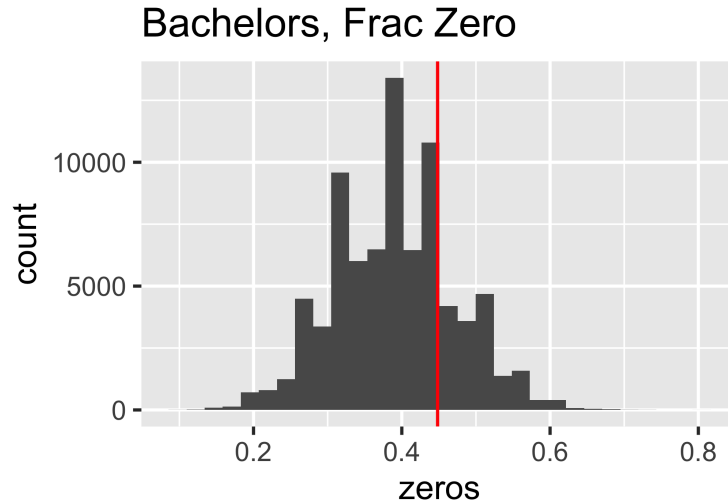
All 4 chains finished successfully.

Mean chain execution time: 2.1 seconds.

Total execution time: 8.5 seconds.

| variable | mean    | median  | sd   | mad  | q5      | q95     | rhat | ess_bulk | ess_tail |
|----------|---------|---------|------|------|---------|---------|------|----------|----------|
| lp__     | -261.16 | -260.85 | 1.02 | 0.71 | -263.21 | -260.21 | 1.00 | 35823    | 47636    |
| lambda1  | 0.95    | 0.94    | 0.13 | 0.13 | 0.75    | 1.17    | 1.00 | 75976    | 54495    |
| lambda2  | 1.40    | 1.40    | 0.08 | 0.08 | 1.27    | 1.54    | 1.00 | 75256    | 53473    |
| y1rep[1] | 0.95    | 1.00    | 0.98 | 1.48 | 0.00    | 3.00    | 1.00 | 80952    | 80389    |
| y1rep[2] | 0.95    | 1.00    | 0.98 | 1.48 | 0.00    | 3.00    | 1.00 | 79270    | 79427    |
| y1rep[3] | 0.95    | 1.00    | 0.98 | 1.48 | 0.00    | 3.00    | 1.00 | 79299    | 78493    |
| y1rep[4] | 0.95    | 1.00    | 0.98 | 1.48 | 0.00    | 3.00    | 1.00 | 78656    | 80525    |
| y1rep[5] | 0.95    | 1.00    | 0.98 | 1.48 | 0.00    | 3.00    | 1.00 | 78714    | 78498    |
| y1rep[6] | 0.96    | 1.00    | 0.99 | 1.48 | 0.00    | 3.00    | 1.00 | 80345    | 79532    |
| y1rep[7] | 0.94    | 1.00    | 0.98 | 1.48 | 0.00    | 3.00    | 1.00 | 79840    | 79077    |

# Predictive Checks (Pois. Model)





# ZIP

Running MCMC with 4 sequential chains...

Chain 1 finished in 0.4 seconds.

Chain 2 finished in 0.4 seconds.

Chain 3 finished in 0.5 seconds.

Chain 4 finished in 0.5 seconds.

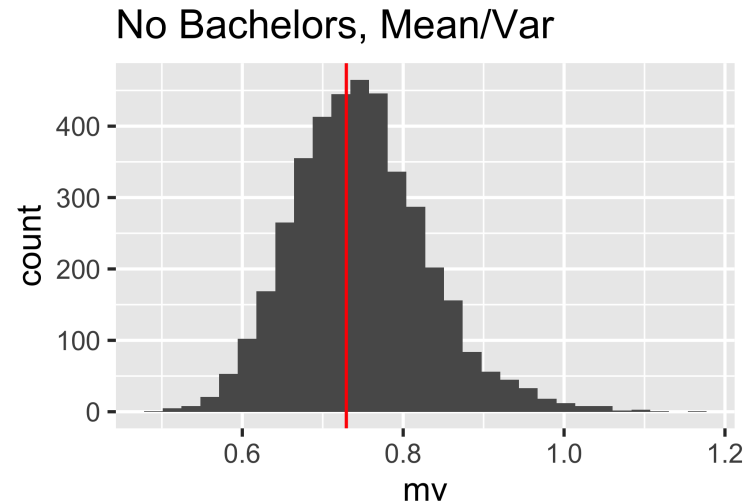
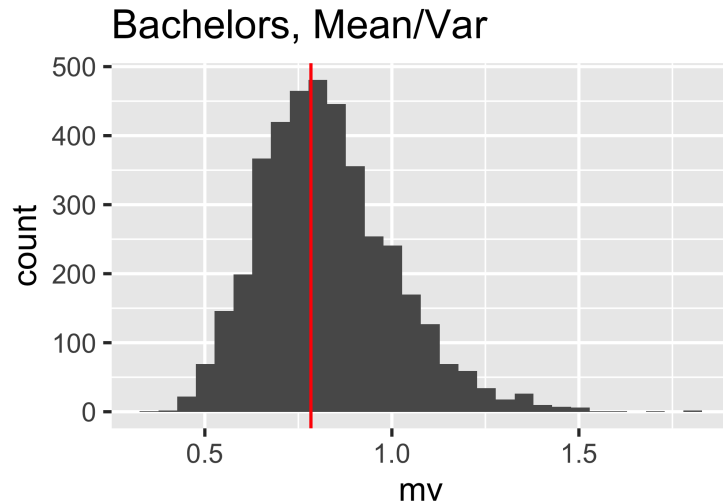
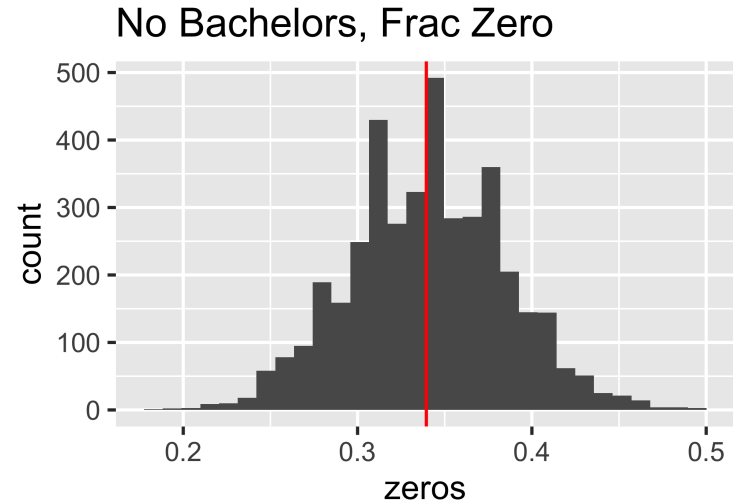
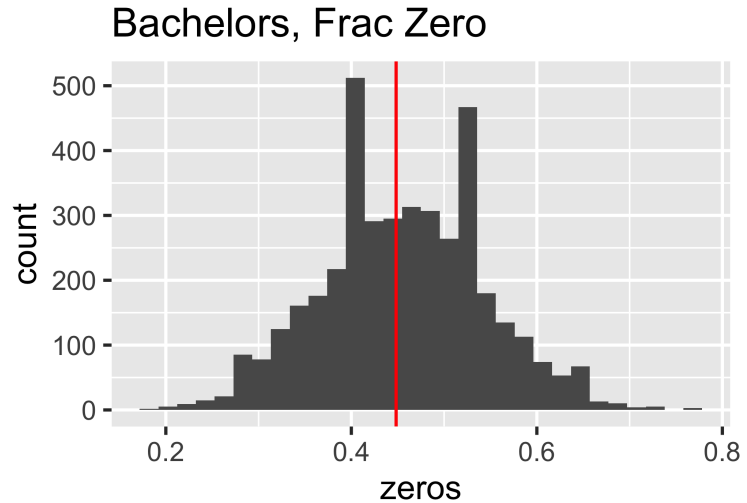
All 4 chains finished successfully.

Mean chain execution time: 0.5 seconds.

Total execution time: 2.0 seconds.

| variable | mean    | median  | sd   | mad  | q5      | q95     | rhat | ess_bulk | ess_tail |
|----------|---------|---------|------|------|---------|---------|------|----------|----------|
| lp__     | -423.61 | -423.24 | 1.54 | 1.28 | -426.63 | -421.84 | 1.00 | 1454     | 2020     |
| lambda1  | 1.20    | 1.18    | 0.21 | 0.21 | 0.89    | 1.58    | 1.00 | 2372     | 2583     |
| lambda2  | 1.75    | 1.75    | 0.13 | 0.13 | 1.54    | 1.96    | 1.00 | 2604     | 2453     |
| theta1   | 0.21    | 0.21    | 0.10 | 0.11 | 0.04    | 0.38    | 1.00 | 2053     | 1306     |
| theta2   | 0.20    | 0.20    | 0.04 | 0.04 | 0.13    | 0.27    | 1.00 | 2503     | 1935     |
| ylrep[1] | 0.94    | 1.00    | 1.12 | 1.48 | 0.00    | 3.00    | 1.00 | 3939     | 3904     |
| ylrep[2] | 0.96    | 1.00    | 1.11 | 1.48 | 0.00    | 3.00    | 1.00 | 3790     | 3780     |
| ylrep[3] | 0.94    | 1.00    | 1.10 | 1.48 | 0.00    | 3.00    | 1.00 | 3330     | 3069     |
| ylrep[4] | 0.92    | 1.00    | 1.09 | 1.48 | 0.00    | 3.00    | 1.00 | 3215     | 3372     |
| ylrep[5] | 0.94    | 1.00    | 1.10 | 1.48 | 0.00    | 3.00    | 1.00 | 3984     | 3862     |

# Predictive Checks (ZIP Model)



# Model Comparison

- PPCs for checking model fit. But how do we compare different models (regardless of fit)?
- Compare models by their out-of-sample predictive power
- Leave-one-out cross validation
  - Fit model  $n$  times, leaving out one observation
  - Evaluate predictive accuracy on left out observations
  - Typically used for parameter tuning
- Can use ideas from LOO-CV to compare models

# Measures of Predictive Accuracy

- Point prediction: predict a single unknown future observation
- Measures of predictive accuracy are called *scoring functions* (e.g. MSE)
- Probabilistic predictions account for uncertainty
  - **Scoring rules:** measure of accuracy based on probabilistic predictions
  - Examples: quadratic, logarithmic and zero-one scores

# Log-likelihood as a scoring rule

- As a measure of accuracy, we call it the “log predictive density”
- $\log(p(y|\theta))$  is proportional to MSE if the data are normal with constant variance
- Connection to Kullback-Leibler divergence
  - KL:  $-E_p \log(q(y)/p(y)) = -E_p \log(q(y)) + E_p \log(p(y))$
  - Asymptotically, the lowest KL model is the one with the highest expected log predictive density

# Out-of-sample prediction

# Expected Log Pointwise Predictive Density

# Pareto-Smoothed Importance Sampling

- When the data are conditionally independent,

$r_i^s = \frac{1}{p(y_i|\theta^s)} \propto \frac{p(\theta^s|y_{-i})}{p(\theta^s|y)}$  to get importance sampling estimates

- $p(\tilde{y}_i | y_{-i}) \approx \frac{\sum_{s=1}^S r_i^s p(\tilde{y}_i | \theta^s)}{\sum_{s=1}^S r_i^s}$
- $p(y_i | y_{-i}) \approx \frac{1}{\frac{1}{S} \sum_{s=1}^S \frac{1}{p(y_i|\theta^s)}}$

<https://link.springer.com/article/10.1007/s11222-016-9696-4>



```

1 library(loo)
2 nb_model <- cmdstan_model("nb_model.stan")
3 nb_fit = nb_model$sample(
4     data=list(n1=num_bachelors, n2=num_no_bachelors,
5     y1=bachelors_data, y2=no_bachelors_data),
6     refresh=0)

```

Running MCMC with 4 sequential chains...

Chain 1 finished in 0.4 seconds.  
Chain 2 finished in 0.4 seconds.  
Chain 3 finished in 0.4 seconds.  
Chain 4 finished in 0.5 seconds.

All 4 chains finished successfully.  
Mean chain execution time: 0.4 seconds.  
Total execution time: 1.8 seconds.

```

1 loo_compare(list("pois"=pois_fit$loo(),
2     "zip"=zip_fit$loo(),
3     "nb"=nb_fit$loo()))

```

|      | elpd_diff | se_diff |
|------|-----------|---------|
| zip  | 0.0       | 0.0     |
| nb   | -3.2      | 2.4     |
| pois | -8.6      | 4.6     |

# Galaxies Example

- Velocities in km/sec of 82 galaxies (Corona Borealis region).
- Multimodality in such surveys is evidence for voids and superclusters in the far universe.
- Statistical question: how many clusters are there in this dataset?

# Model Comparison

```
1 library("loo")
2 galaxy_speeds <- log(MASS::galaxies)
3 galaxy_results <- list()
4
5 ## Run model for each of k clusters
6 mix_model <- cmdstanr::cmdstan_model("mix_model.stan")
7 for(k in 1:5) {
8   galaxy_results[[k]] <- mix_model$sample(data=list(K=k, N=length(galaxy_sp
9                                           refresh=0, show_messages=FALSE)
10 }
```

Warning: 63 of 4000 (2.0%) transitions ended with a divergence.

See <https://mc-stan.org/misc/warnings> for details.

Warning: 16 of 4000 (0.0%) transitions hit the maximum treedepth limit of 10.

See <https://mc-stan.org/misc/warnings> for details.

# Model Comparison

```
1 loo_compare(galaxy_results[[1]]$loo(),  
2             galaxy_results[[2]]$loo(),  
3             galaxy_results[[3]]$loo(),  
4             galaxy_results[[4]]$loo(),  
5             galaxy_results[[5]]$loo())
```

|        | elpd_diff | se_diff |
|--------|-----------|---------|
| model3 | 0.0       | 0.0     |
| model5 | -0.4      | 1.4     |
| model4 | -1.0      | 0.7     |
| model2 | -10.7     | 3.7     |
| model1 | -41.1     | 8.7     |